CA3411 PROJECT WORK

MCA
IV Semester / II Year

AI BASED TWO WHEELERS SAFETY MONITOR SYSTEM

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INTRODUCTION

- Two-wheelers are widely used but riders are highly vulnerable to accidents.
- Major causes include lack of helmet usage, poor awareness, and delayed emergency response.
- Existing systems rely on manual monitoring and lack real-time safety mechanisms.
- This project proposes an AI-based system using Computer Vision and IoT for continuous safety monitoring.
- The system detects accidents, monitors conditions, and sends alerts to improve rider safety.

BASE PAPER

Title of Paper & Year	Author / Publication	Methodology	Pros	Cons / Gap
An IoT-Based Vehicle Accident Detection and Classification System Using Sensor Fusion (2023)	Cheng et al. (arXiv)	• YOLOv8 integrated with Transformer blocks for improved feature extraction • Detects helmet vs no-helmet using real-time video frames • Handles small objects, occlusion, and complex backgrounds	• High detection accuracy in real-time scenarios • Robust performance in complex traffic conditions • Efficient for real-time monitoring systems	• No accident detection or emergency response • Does not monitor surrounding environment • Lacks integration with IoT or sensor systems
IoT-Based Accident Detection System (2022)	Al Mamun et al.	• Sensor fusion using accelerometer and gyroscope data • Real-time monitoring of vehicle movement	• Reduces false positives compared to single sensor systems • Enables real-time emergency alert triggering • Suitable for IoT-based implementations	• No visual confirmation (camera not used) • Cannot verify accident context properly • Purely reactive system (no prevention)

SUMMARY OF LITERATURE SURVEY

- Existing works mainly focus on **helmet detection, accident detection, and rear monitoring** as separate systems
- Deep learning models (YOLO) improve **helmet detection accuracy**, but lack integration with other safety features
- Sensor-based approaches using accelerometer and gyroscope enable **real-time accident detection**, but lack context awareness
- Rear monitoring systems provide **basic proximity alerts**, but are limited in accuracy under real-world conditions
- Overall, most systems are **reactive and isolated**, highlighting the need for an integrated and proactive safety solution

OBJECTIVES

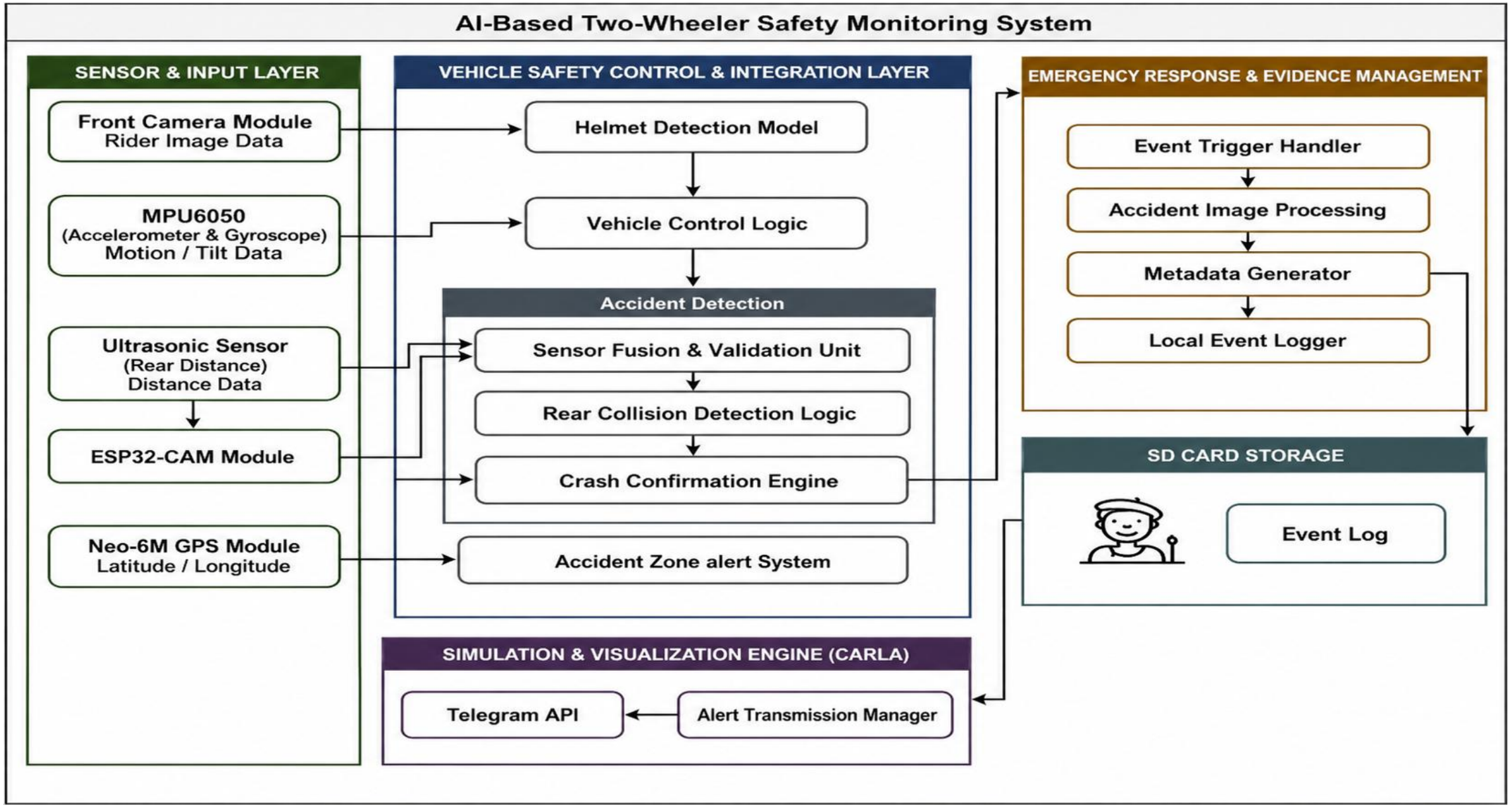
01 To develop a system for **real-time helmet detection** using computer vision techniques to ensure rider safety compliance.

02 To design and implement an accident detection mechanism using accelerometer and gyroscope sensors for identifying abnormal vehicle movements

03 To build a rear-side monitoring system using ultrasonic sensors and camera to detect approaching vehicles and prevent collisions..

04 To implement an emergency alert system that automatically sends location and evidence to predefined contacts without smartphone dependency.

SYSTEM ARCHITECTURE



Modules

Module 1: Helmet Detection Module

- Detects whether the rider is wearing a helmet using a front camera
- Controls vehicle operation based on detection result

Module 2: Accident Detection & Emergency Alert Module

- Detects accidents using motion sensor data (accelerometer & gyroscope)
- Sends emergency alerts with location during critical events

Module 3: Rear Accident / Rear-Side Detection Module

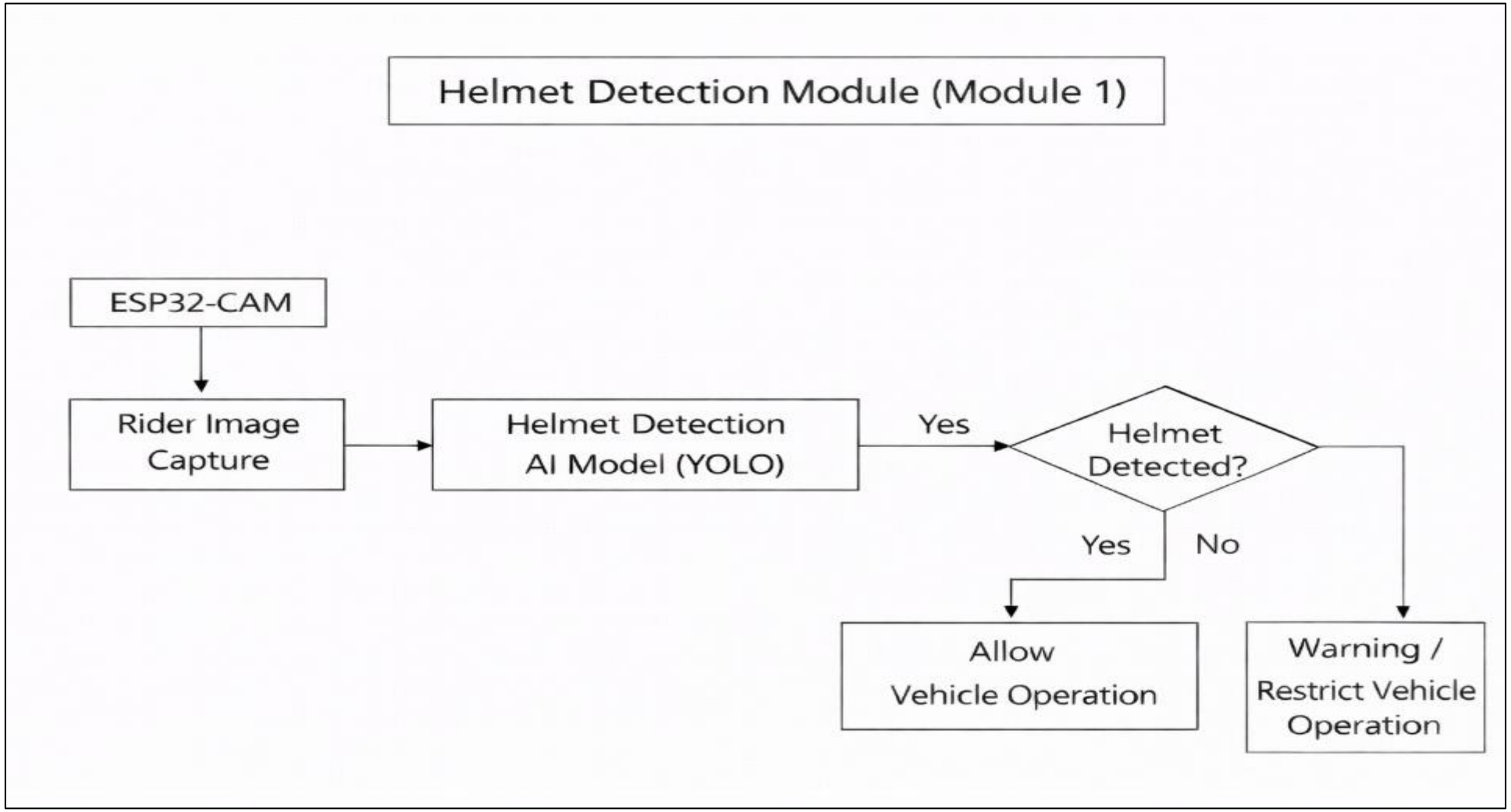
- Monitors rear-side traffic using camera and proximity sensors
- Provides warnings for approaching vehicles and collision risk

Module 4: Accident Zone Warning Module

- Identifies accident-prone areas using DBSCAN clustering on GPS data
- Alerts the rider when entering high-risk zones

"If obstacle, take left"

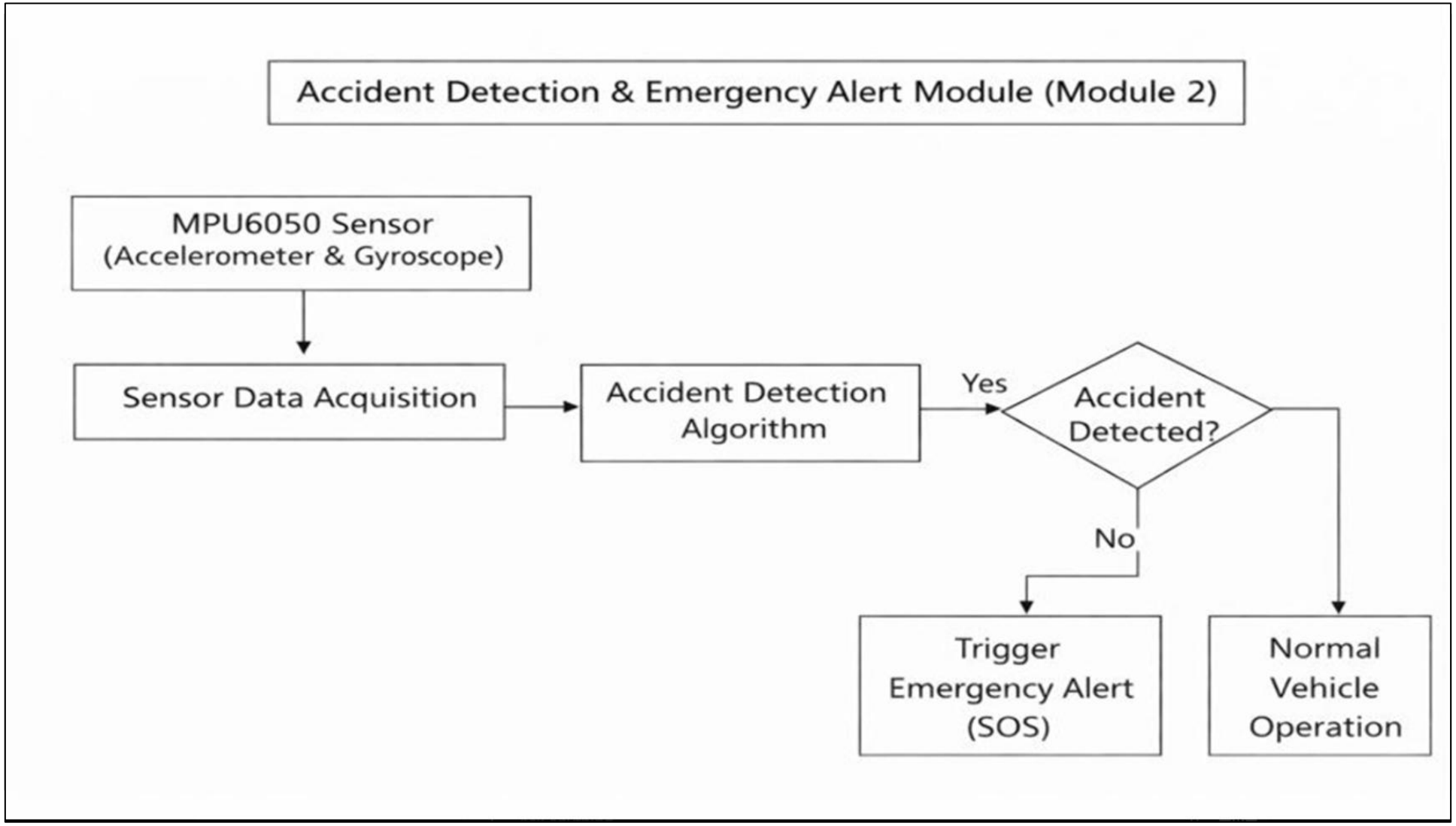
MODULE 1 FLOW DIAGRAM



Module 1 - Helmet Detection Module

- Detects whether the rider is wearing a helmet using a front-mounted camera during the ride
- Uses a deep learning-based model (YOLOv8) to analyze captured frames in real time
- Classifies the rider into helmet or no-helmet category based on detection results
- Regulates vehicle operation by applying restrictions when helmet is not detected
- Ensures continuous monitoring to improve rider safety compliance

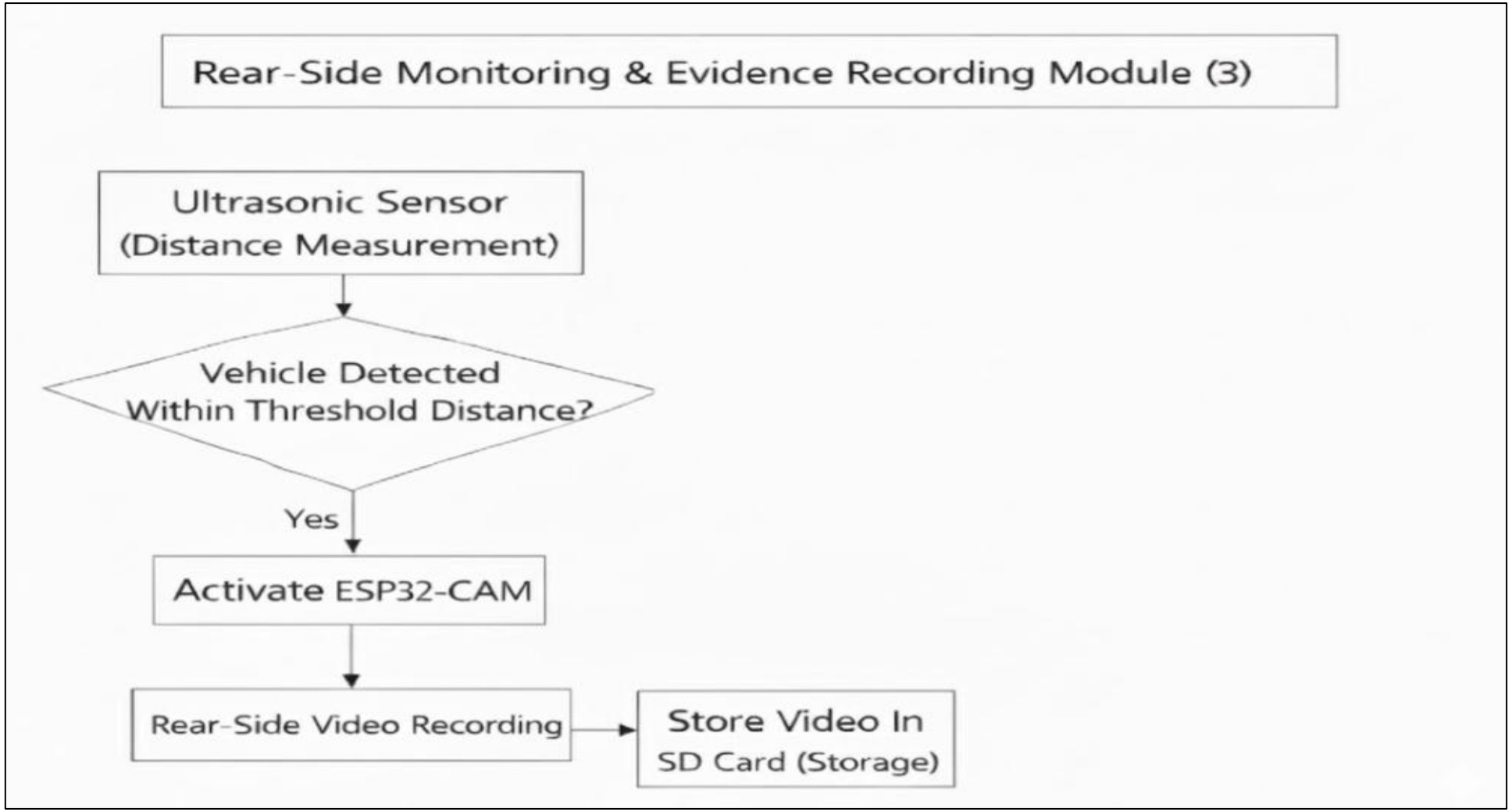
MODULE 2 FLOW DIAGRAM



Module 2 - Accident Detection & Emergency Alert

- Detects accidents by analyzing motion data from accelerometer and gyroscope sensors
- Identifies sudden falls, abnormal tilt, or impact conditions indicating a crash event
- Uses validation logic to confirm genuine accidents and reduce false alerts
- Captures accident images and stores them as evidence for further analysis
- Sends emergency alert with location and details to predefined contacts automatically

MODULE 3 FLOW DIAGRAM



Module 3 – Rear Accident Detection

- Monitors rear-side traffic conditions using ultrasonic sensor and camera system
- Continuously measures distance between vehicle and approaching objects from behind
- Classifies distance into safe, warning, and critical zones for risk evaluation
- Generates alerts when vehicles enter danger zone to avoid rear-end collision
- Records rear-side events for evidence and improves situational awareness

Module 4 – Accident Zone Warning Module

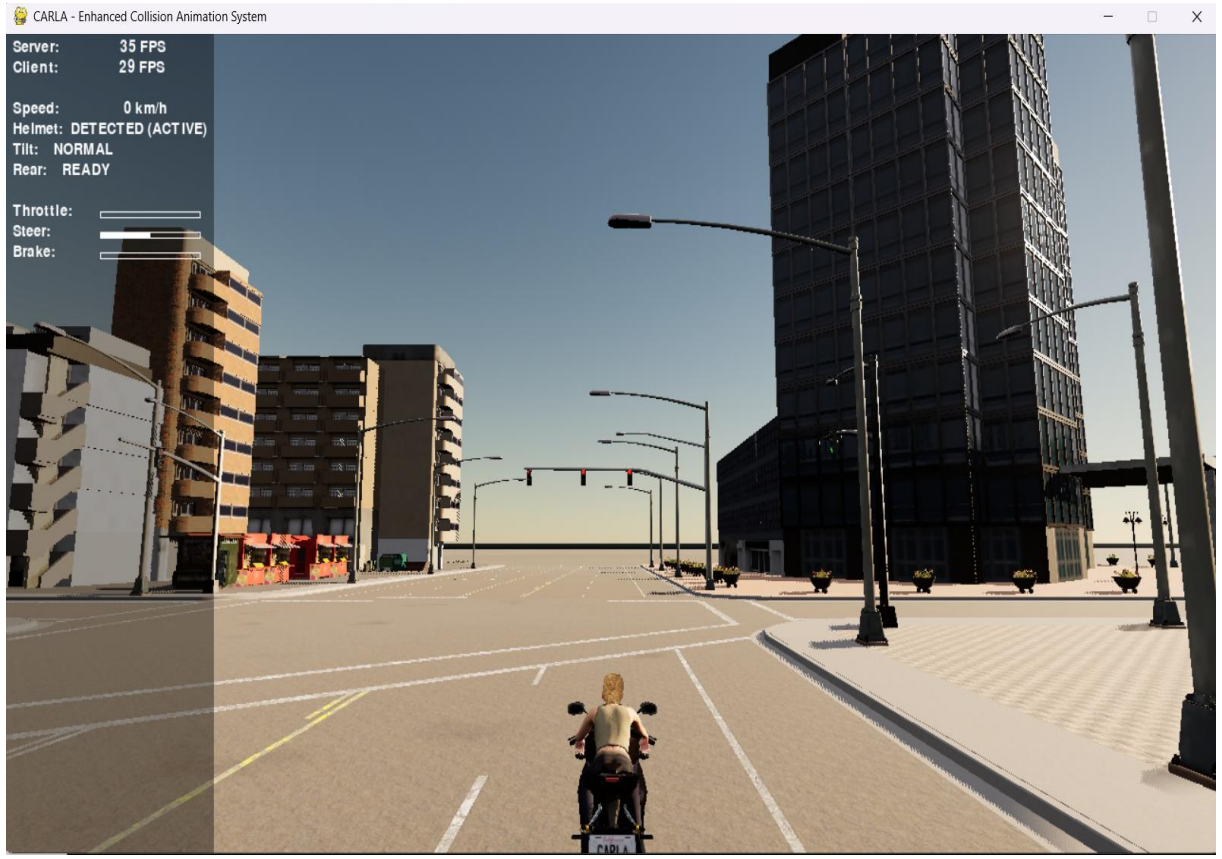
- Uses DBSCAN clustering algorithm on historical accident GPS data to identify hotspots
- Detects accident-prone zones based on density of past accident occurrences
- Continuously tracks rider’s real-time location using GPS module
- Calculates distance between rider and hotspot using Haversine formula
- Provides early warning alerts when entering high-risk areas to prevent accidents

RESULTS

Helmet Detection Module

```
--- Checking helmet status ---
√ HELMET DETECTED - With Helmet @ 0.89
[DEBUG] Read from fall_status.txt: 'NONE'
√ Tilt Status: NONE (Normal)

--- Checking helmet status ---
√ HELMET DETECTED - With Helmet @ 0.90
[DEBUG] Read from fall_status.txt: 'NONE'
√ Tilt Status: NONE (Normal)

--- Checking helmet status ---
√ HELMET DETECTED - With Helmet @ 0.87
[DEBUG] Read from fall_status.txt: 'NONE'
√ Tilt Status: NONE (Normal)

--- Checking helmet status ---
√ HELMET DETECTED - With Helmet @ 0.88
[DEBUG] Read from fall_status.txt: 'NONE'
√ Tilt Status: NONE (Normal)

--- Checking helmet status ---
√ HELMET DETECTED - With Helmet @ 0.78
[DEBUG] Read from fall_status.txt: 'NONE'
√ Tilt Status: NONE (Normal)

--- Checking helmet status ---
√ HELMET DETECTED - With Helmet @ 0.68
[DEBUG] Read from fall_status.txt: 'NONE'
√ Tilt Status: NONE (Normal)
```

Helmet Detection output in Terminal



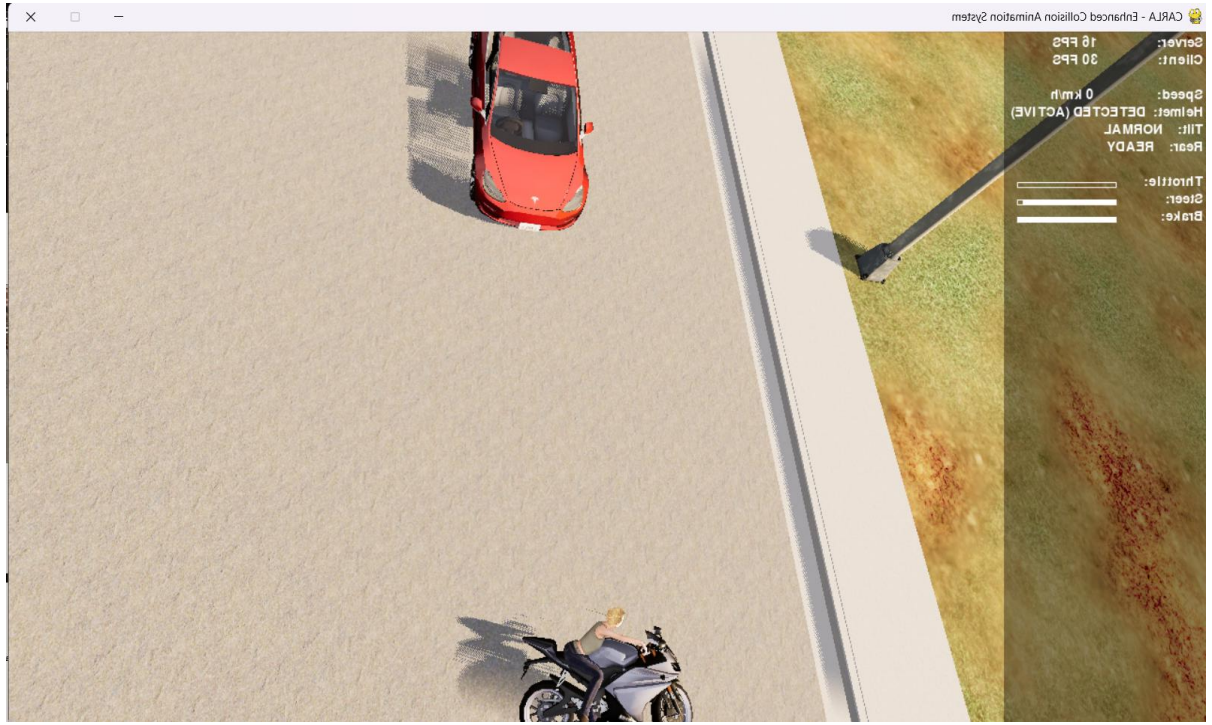
Helmet Detection Output in CARLA

RESULTS

Accident Detection & Emergency Alert

```
============================================================
FALL DETECTION SYSTEM - Enhanced Tilt Sensitivity
============================================================
Serial Port       : COM20
Baud Rate         : 9600
Output File       : d:\CEG-MCA\Semester-4\Final-Year-Project\
Stability Count   : 3 readings
Debounce Time     : 0.1s
============================================================
✓ Connected to COM20

[10:51:49.704] ✓ State Updated: NONE
Monitoring tilt sensor... (Press Ctrl+C to stop)

[10:51:50] → Detecting: LEFT (count: 1/3)
[10:51:50.924] ✓ State Updated: LEFT
[10:51:52.430] ✓ State Updated: LEFT
[10:51:53.929] ✓ State Updated: LEFT
[10:51:55.436] ✓ State Updated: LEFT
[10:51:56.936] ✓ State Updated: LEFT
[10:51:58.438] ✓ State Updated: LEFT
```

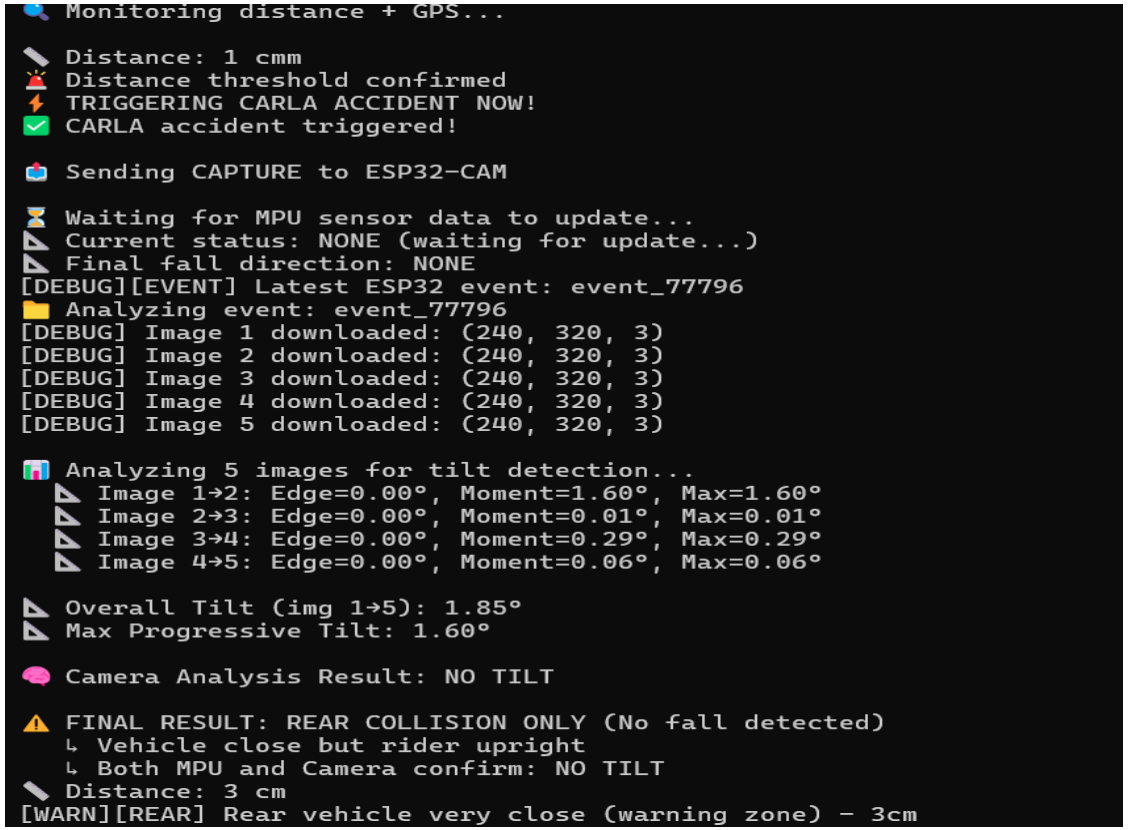
Fall Detection output in Terminal



Fall Detection output in CARLA

RESULTS

Rear Accident and Detection Module:-



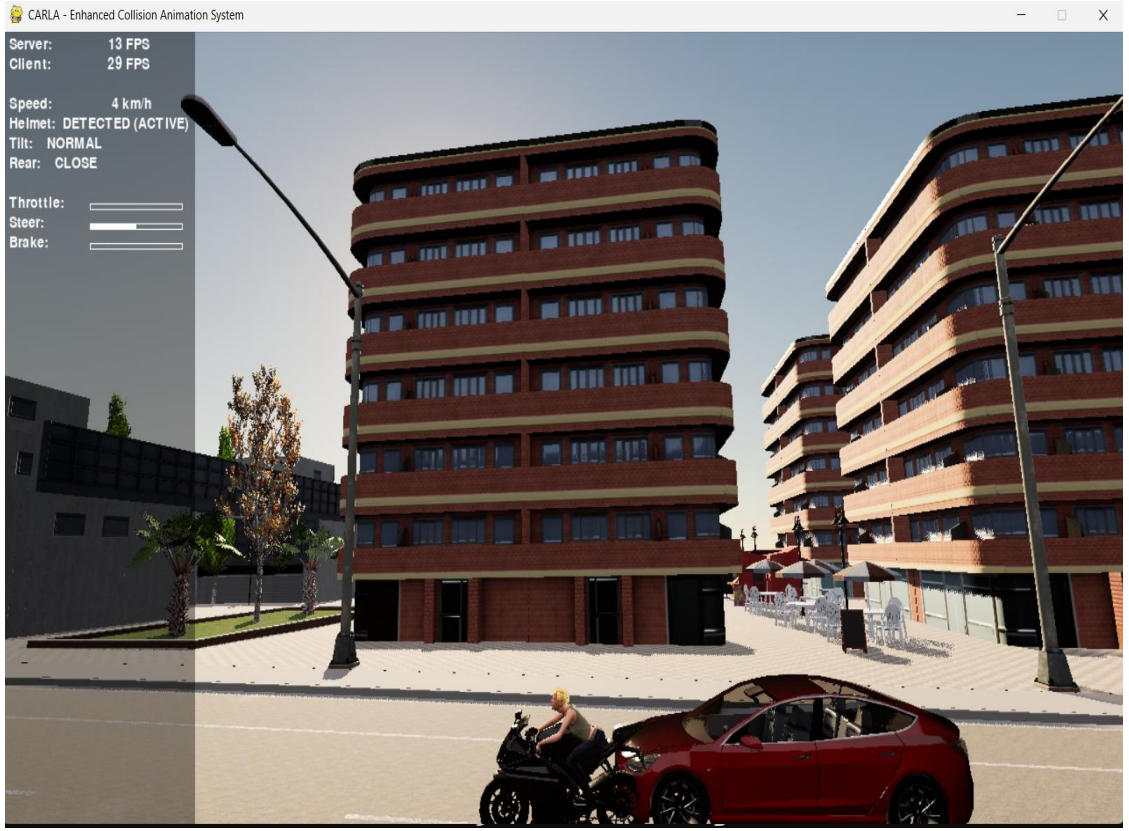
```
🔍 Monitoring distance + GPS...

📏 Distance: 1 cmm
🚨 Distance threshold confirmed
⚡ TRIGGERING CARLA ACCIDENT NOW!
✅ CARLA accident triggered!

📤 Sending CAPTURE to ESP32-CAM

⏳ Waiting for MPU sensor data to update...
📐 Current status: NONE (waiting for update...)
📐 Final fall direction: NONE
[DEBUG][EVENT] Latest ESP32 event: event_77796
📁 Analyzing event: event_77796
[DEBUG] Image 1 downloaded: (240, 320, 3)
[DEBUG] Image 2 downloaded: (240, 320, 3)
[DEBUG] Image 3 downloaded: (240, 320, 3)
[DEBUG] Image 4 downloaded: (240, 320, 3)
[DEBUG] Image 5 downloaded: (240, 320, 3)

📊 Analyzing 5 images for tilt detection...
  📐 Image 1→2: Edge=0.00°, Moment=1.60°, Max=1.60°
  📐 Image 2→3: Edge=0.00°, Moment=0.01°, Max=0.01°
  📐 Image 3→4: Edge=0.00°, Moment=0.29°, Max=0.29°
  📐 Image 4→5: Edge=0.00°, Moment=0.06°, Max=0.06°

📐 Overall Tilt (img 1→5): 1.85°
📐 Max Progressive Tilt: 1.60°

🧠 Camera Analysis Result: NO TILT

⚠️ FINAL RESULT: REAR COLLISION ONLY (No fall detected)
  ↳ Vehicle close but rider upright
  ↳ Both MPU and Camera confirm: NO TILT
📏 Distance: 3 cm
[WARN][REAR] Rear vehicle very close (warning zone) - 3cm
```

Rear Collision Detection in Terminal

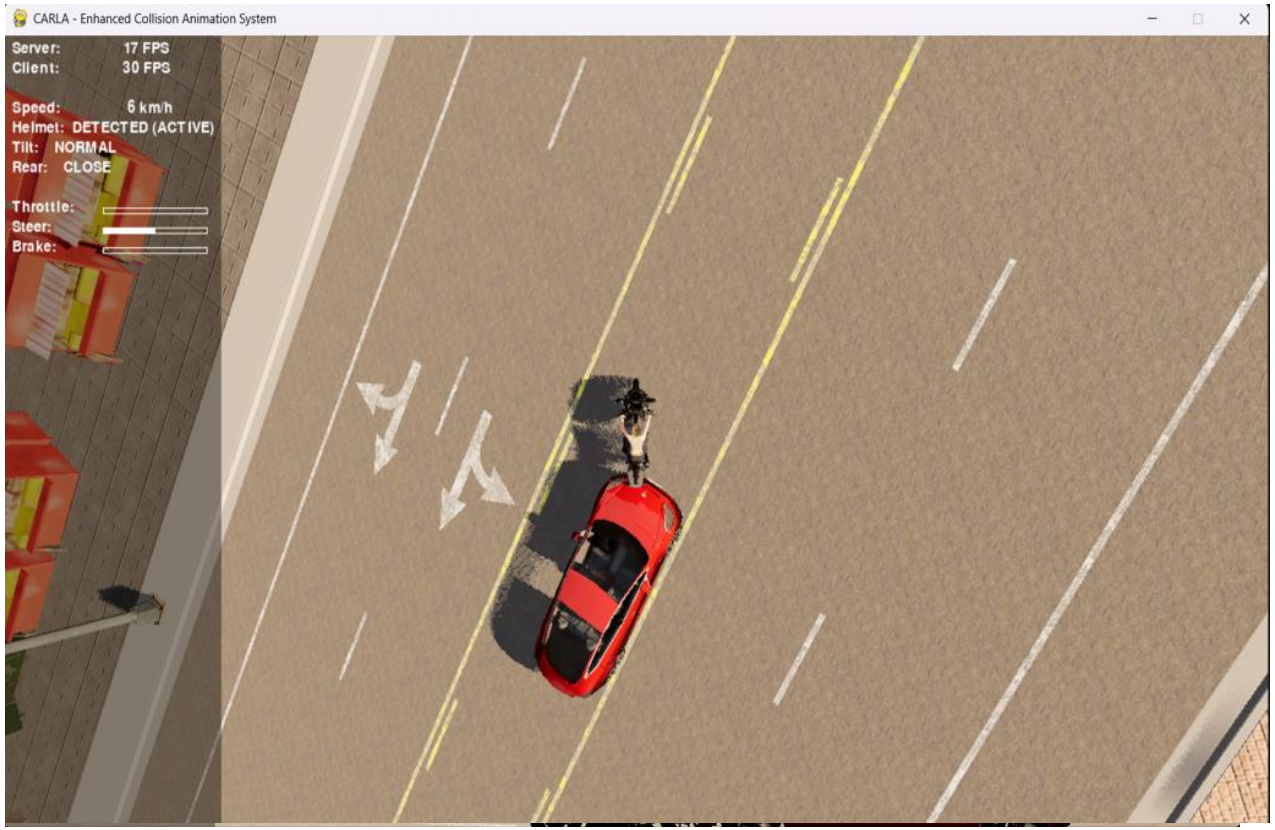
Rear Proximity Alert for Approaching Vehicle in CARLA

RESULTS

Rear Accident and Detection Module:-



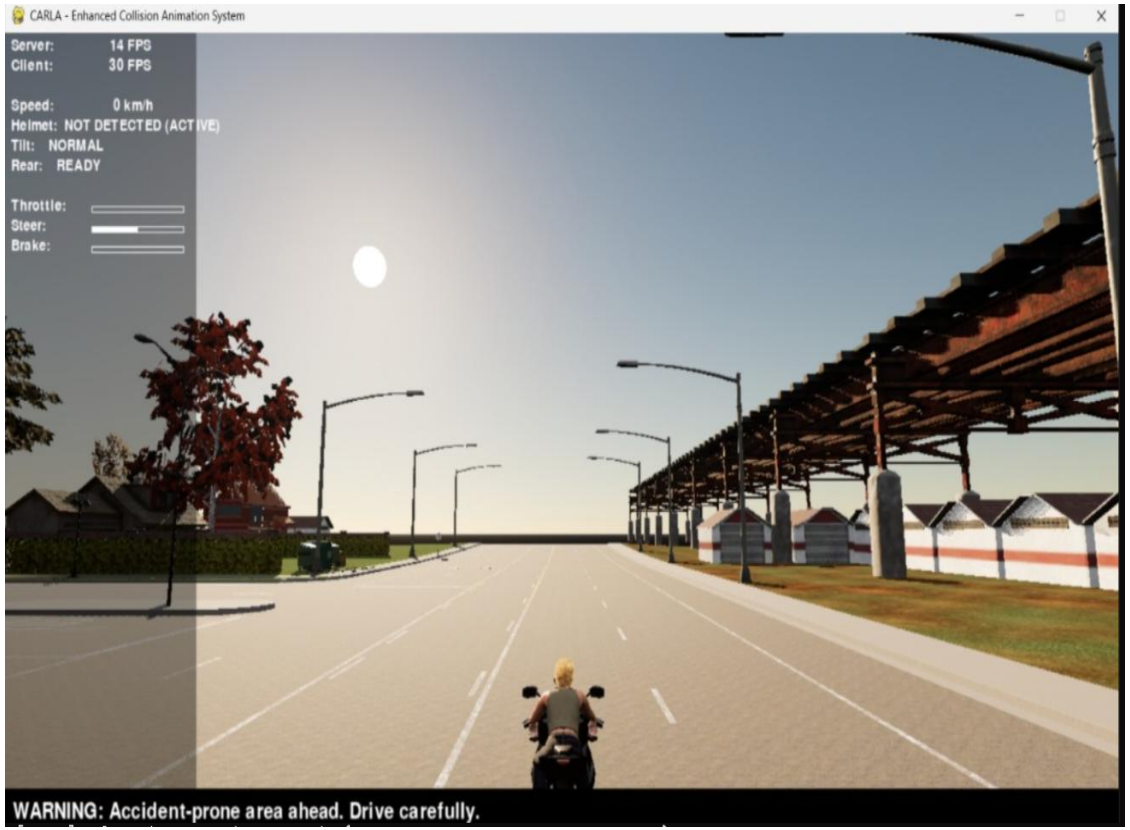
Rear Collision in CARLA Simulator



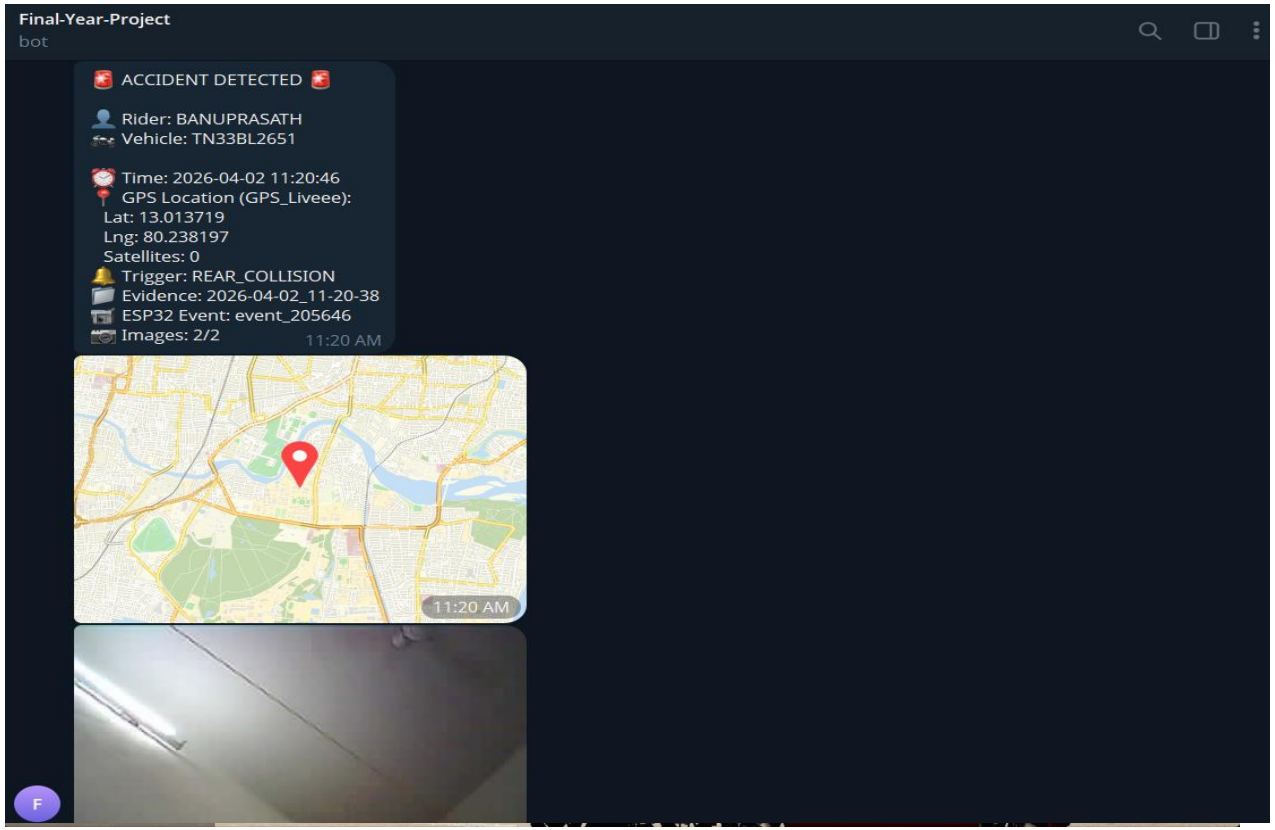
Rear Collision in CARLA simulator Top View

RESULTS

Accident Zone Warning Module:-

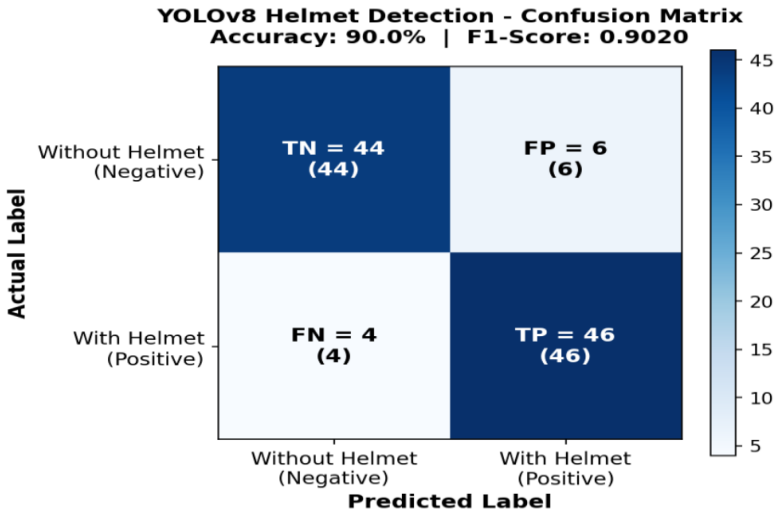


Accident Zone warning in CARLA Simulator



Telegram message with Location and Evidence

PERFORMANCE ANALYSIS



Confusion Matrix for Helmet Detection

Metric	Value
Threshold	30°
Success Rate	92%
False Positives	5%
Response Time	5 sec

Tilt Detection Performance

Metric	Value
Accuracy	95%
False Positives	3%
Latency	300 ms

Rear Collision Detection Performance

Module	Metric	Value
Helmet Detection	mAP	88%
Tilt Detection	Success Rate	92%
Rear Detection	Accuracy	95%
Emergency Alert	Time	4–8s

Overall System Performance

CONCLUSION

- Successfully developed an AI-based integrated two-wheeler safety monitoring system combining computer vision, IoT sensors, and real-time processing.
- The system ensures rider safety through accurate helmet detection using YOLOv8 and continuous compliance monitoring.
- Accident detection using motion sensors provides reliable crash identification with reduced false positives.
- DBSCAN-based accident zone prediction enables proactive safety by providing early warnings in high-risk areas.
- The emergency alert mechanism ensures timely communication with location and evidence, improving response during critical situations.

REFERENCES

[1] A. M. Addarani, E. Rohadi, and M. Sarosa, “Analysis of Driving Safety with Distance Detection Systems for Motorized Vehicles Using Ultrasonic Sensors,” in Proc. Int. Seminar on Intelligent Technology and Its Applications (ISITIA), 2023, pp.

[2] An IoT-Based Vehicle Accident Detection and Classification System Using Sensor Fusion, IEEE Journal, 2023.

[3] MotoWatch: Deep Learning Based AI Assisted Helmet Wearing Detection System, IEEE International Conference on Intelligent Computing and Control Systems (ICICCS), 2025.

[4] Tiny-YOLOv8 Long-Distance Electric Vehicle Helmet Detection Model, IEEE 5th International Conference on Computer Vision, Application and Algorithm (CVAA), 2025.

[5] Real-Time Helmet Detection System with Vehicle Number Extraction for Two-Wheeler Vehicles Using YOLOv8, IEEE Conference Proceedings, 2024.

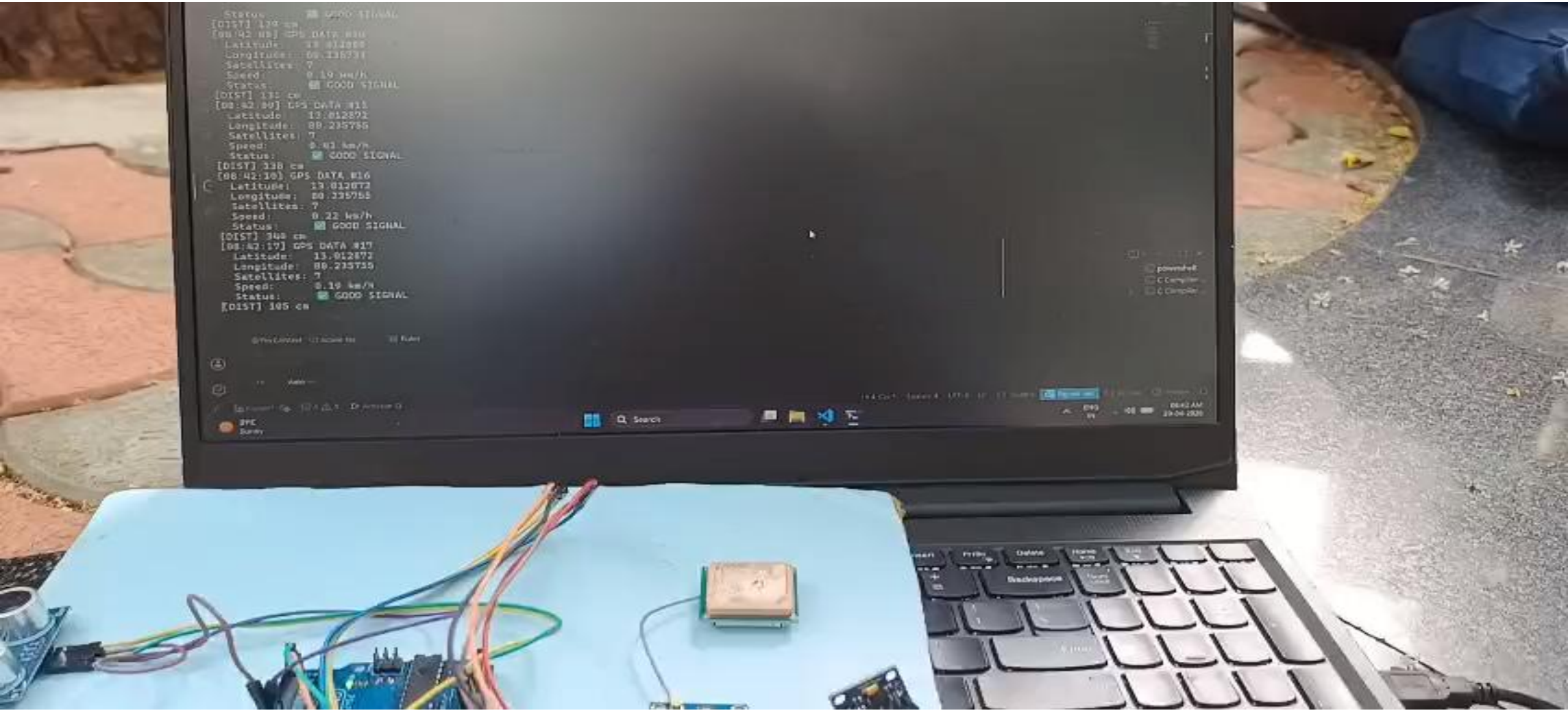
[6] Smart Helmet with Automatic Braking Using IoT, IEEE 2nd International Conference on IoT, Communication and Automation Technology (ICICAT), 2024.

Thank You

CARLA Simulation Demo

Demo crash Detection

Demo of Neo 6M



PICTORIAL REPRESENTATION